
UNIT 7 DATA MINING – AN INTRODUCTION

Structure

- 7.0 Introduction
- 7.1 Objectives
- 7.2 Data Mining and its Benefits
- 7.3 Purpose Of Data Mining
- 7.4 How Does Data Mining Works?
- 7.5 Data Mining Techniques
- 7.6 Data Mining Vs Data Warehousing
- 7.7 Data Mining Tools
- 7.8 Applications of Data Mining
- 7.9 Issues in Data Mining
- 7.10 Summary
- 7.11 Solutions / Answers
- 7.12 Further Readings

7.0 INTRODUCTION

We are in an age often referred to as the information age in which we have been collecting huge amounts of data. Thanks to sophisticated technologies such as computers, satellites, etc. With the invention of computers and mass digital storage media, we started collecting and storing all sorts of data. Unfortunately, these huge data stored on different storage media became profuse. This chaotic situation of storing the data has led to the creation of database management systems (DBMS). The boosting of database management systems also contributes to massive gathering of data of all types. Today, we have a lot of information than we can handle - from satellite pictures, text reports to business transactions, scientific data, military intelligence, medical and personal data, games, digital media etc.. Information retrieval is simply not enough anymore for decision-making. Opposed with huge collections of data, we have generated new requirements for helping us to make better managerial decisions. These requirements may be automatic summarization of data, extraction of the kernel of the information stored, and the discovery of patterns in raw data.

In the earlier 2 blocks of this course, we had studied about the data warehousing and related concepts. In this unit, we will go through why and how to mine data from huge data sources.

Data Mining is a technical process by which consistent patterns are identified, explored, sorted and organized. Data Mining is used to provide two primary advantages – (i) to give businesses the predictive power to estimate the unknown or future values and (ii) to provide businesses the descriptive power by finding interesting patterns in the data.

With data mining tools, organizations of any size can extract valuable insights from their datasets, including information about consumers, costs, and future trends. This process can be employed to (a) answer business questions that were traditionally too time-consuming to address and (b) make knowledge-driven decisions based on the absolute best data available.

7.1 OBJECTIVES

After going through this unit, you should be able to:

- understand the definition of data mining;
- identify the purpose of Text Mining;
- describe process of data mining;
- narrate the key differences between the data mining and data warehousing;
- list and discuss various data mining tools, and
- discuss various data mining applications.

7.2 DATA MINING AND ITS BENEFITS

Data mining which is also known as knowledge discovery in data (KDD) is the process of uncovering patterns and other valuable information from large data sets. Given the evolution of data warehousing technology and the growth of big data, adoption of data mining techniques has rapidly accelerated over the last couple of decades, assisting companies by transforming their raw data into useful knowledge. However, despite the fact that that technology continuously evolves to handle data at a large-scale, leaders still face challenges with scalability and automation.

Data mining has improved organizational decision-making through insightful data analyses. The data mining techniques that underpin these analyses can be divided into two main purposes; they can either describe the target dataset or they can predict outcomes through the use of machine learning algorithms. These methods are used to organize and filter data, surfacing the most interesting information, from fraud detection to user behaviors, bottlenecks, and even security breaches.

The purpose of mining the data can be multi-fold which includes:

- Predicting various outcomes;
- Modeling target audience;
- Collecting the information about the product.

With this technique, we analyze the data and then convert that data into meaningful information. This helps the business to take accurate and better decisions in an organization. It helps to develop smart market decision, run accurate campaigns, make predictions, and more. With the help of Data mining, we can analyze customer behaviors and their insights. This leads to great success and data-driven business.

In order to have a thorough understanding of the behavior of the customers, Data mining helps to extract the pattern from the databases. The knowledge thus acquired, allows the companies to offer the best possible services.

7.2.1 Benefits of Data Mining

Data mining provides us with the means of resolving problems in this challenging information age. Data mining benefits include:

- It helps companies gather reliable information.
- It's an efficient, cost-effective solution compared to other data applications.

- It helps businesses make profitable production and operational adjustments.
- Data mining uses both new and legacy systems.
- It helps businesses make informed decisions.
- It helps detect credit risks and fraud.
- It helps data scientists easily analyze enormous amounts of data quickly.
- Data scientists can use the information to detect fraud, build risk models and improve product safety
- It helps data scientists quickly initiate automated predictions of behaviors and trends and discover hidden patterns.

7.2.2 Types of Data that can be Mined

In principle, data mining is not specific to one type of media or data. Data mining should be applicable to any kind of information repository. However, algorithms and approaches may differ when applied to different types of data. Indeed, the challenges presented by different types of data vary significantly.

Data mining is being put into use and studied for databases, including relational databases, object-relational databases and object-oriented databases, data warehouses, transactional databases, unstructured and semi-structured repositories such as the World Wide Web, social media data, advanced databases such as spatial databases, multimedia databases, time-series databases and textual databases, and even flat files.

7.3 HOW DOES DATA MINING WORKS?

Cross-Industry Standard Process for Data Mining (CRISP-DM), is an industry-proven way to guide your data mining efforts. As a methodology, it includes descriptions of the typical phases of a project, the tasks involved with each phase, and an explanation of the relationships between these tasks. As a process model, it provides an overview of the data mining life cycle.

The CRISP-DM model is flexible and can be customized easily. The life cycle model consists of six phases with arrows indicating the most important and frequent dependencies between phases.

The sequence of the phases is not strict. In fact, most projects move back and forth between phases as necessary. Let's take a closer look at each phase of the CRISP-DM as shown in the Figure 1:



Figure 1: Data Mining Life Cycle

Business Understanding

To get started, first ask these questions:

- What is our objective?
- What problem are we trying to solve?
- What data do we need to solve it?

Without a clear understanding of the proper data to mine, the project can produce errors, inaccurate results, or results that don't answer the correct questions.

Data Understanding

Once the overall objective is determined, proper data needs to be collected. The data must be relevant to subject matter and usually comes from a variety of sources such as sales records, customer surveys, and geo-location data. This phase's goal is to ensure the data correctly encompasses all necessary data sets to address the objective.

Data Preparation

The most time-consuming phase, the preparation phase, consists of three steps: extraction, transformation, and loading — also referred to as ETL. First, data is extracted from various sources and deposited into a staging area. Next, during the transformation step: the data is cleaned, null sets are populated, duplicative data is removed, errors are resolved, and all data is allocated into tables. In the final step, loading, the formatted data is loaded into the database for use.

Modeling

Data modeling addresses the relevant data set and considers the best statistical and mathematical approach to answering the objective question(s). There are a variety of modeling techniques available, such as classification, clustering, and regression analysis etc.. It's also not uncommon to use different models on the same data to address specific objectives.

Evaluation

After the models are built and tested, it's time to evaluate their efficiency in answering the question identified during the business understanding phase. This is a human-driven phase, as the individual running the project must determine whether the model output sufficiently meets their objectives. If not, a different model can be created, or different data can be prepared.

Deployment

Once the data mining model is deemed accurate and successful in answering the objective question, it's time to put it to use. Deployment can occur in the form of a visual presentation or a report sharing insights. It also can lead to action such as generating a new sales strategy or implementing risk-reduction measures.

7.4 CLASSIFICATION OF DATA MINING SYSTEMS

Data mining systems can be categorized according to various criteria among other classification are the following:

- **Classification according to the type of data source mined:** this classification categorizes data mining systems according to the type of data handled such as spatial data, multimedia data, time-series data, text data, World Wide Web, etc.
- **Classification according to the data model drawn on:** this classification categorizes data mining systems based on the data model involved such as relational database, object-oriented database, data warehouse, transactional, etc.
- **Classification according to the king of knowledge discovered:** this classification categorizes data mining systems based on the kind of knowledge discovered or data mining functionalities, such as characterization, discrimination, association, classification, clustering, etc. Some systems tend to be comprehensive systems offering several data mining functionalities together.
- **Classification according to mining techniques used:** Data mining systems employ and provide different techniques. This classification categorizes data mining systems according to the data analysis approach used such as machine learning, neural networks, genetic algorithms, statistics, visualization, database-oriented or data warehouse-oriented, etc. The classification can also take into account the degree of user interaction involved in the data mining process such as query-driven systems, interactive exploratory systems, or autonomous systems. A comprehensive system would provide a wide variety of data mining techniques to fit different situations and options, and offer different degrees of user interaction.

7.5 DATA MINING TECHNIQUES

Data mining is most useful in identifying data patterns and deriving useful business insights from those patterns. To accomplish these tasks, data miners use a variety of techniques to generate different results. Here are five common data mining techniques:

7.5.1 Classification Analysis

With this technique, data points are assigned to groups, or classes, based on a specific question or problem to address. For instance, if a consumer packaged goods company wants to optimize its coupon discount strategy for a specific product, it might review inventory levels, sales data, coupon redemption rates, and consumer behavioral data in order to make the best decision possible.

7.5.2 Association Rule Learning

This function seeks to uncover the relationships between data points; it is used to determine whether a specific action or variable has any traits that can be linked to other actions (e.g., business travelers' room choices and dining habits). A hotelier might use association rule insights to offer room upgrades or food and beverage promotions to attract additional business travelers.

7.5.3 Anomaly or Outlier Detection

In addition to searching for patterns, data mining seeks to uncover unusual data within a set. Anomaly detection is the process of finding data that doesn't conform to the pattern. This process can help find instances of fraud and help retailers learn more about spikes, or declines, in the sales of certain products.

7.5.4 Clustering Analysis

Clustering looks for similarities within a data set, separating data points that share common traits into subsets. This is similar to the classification type of analysis in that it groups data points, but, in clustering analysis, the data is not assigned to previously defined groups. Clustering is useful for defining traits within a data set, such as the segmentation of customers based on purchase behavior, need state, life stage, or likely preferences in marketing communication.

7.5.5 Regression Analysis

Regression analysis is about understanding which factors within a data set are most important, which can be ignored, and how these factors interact. With this technique, data miners are able to validate theories such as "when a lot of snow is predicted, more bread and milk will be sold before the storm." While this seems obvious enough there are a number of variables that need to be verified and quantified for the store manager to make sure enough stock is available. For example, how much is "a lot" of snow? How much is "more milk and bread"? Which types of weather forecasts tend to cause consumer action and how many days before the storm will consumers start buying? What is the relationship between inches of snow, units of bread, and units of milk?

Through regression analysis, specific inventory levels of milk and bread (in units/cases) can be recommended for specific levels of snow forecasted (inches), at specific points in time (days before the storm). In this way, the use of regression analysis maximizes sales, minimizes out-of-stock instances, and helps avoid overstocking which results in product spoilage after the storm.

7.6 DATA MINING Vs DATA WAREHOUSING

A data warehouse is a technique for collecting and managing data from varied sources to provide meaningful business insights. It is a blend of technologies and components which allows the strategic use of data. Data Warehouse is repository

of a large amount of data, which is designed for query and analysis instead of transaction processing. It is a process of transforming data into information and making it available to users for analysis.

On the other hand, data mining is looking for hidden, valid, and potentially useful patterns in huge data sets. Data Mining is all about discovering unsuspected/ previously unknown relationships amongst the data. It is a multi-disciplinary skill that uses machine learning, statistics, AI and database technology. The insights extracted via Data mining can be used for marketing, fraud detection, and scientific discovery, etc..

Table 1: Data Mining Vs Data Warehouse

Data Mining	Data Warehouse
Data mining is the process of analyzing unknown patterns of data.	A data warehouse is database system which is designed for analytical instead of transactional work.
Data mining is a method of comparing large amounts of data to finding right patterns.	Data warehousing is a method of centralizing data from different sources into one common repository.
Data mining is usually done by business users with the assistance of engineers.	Data warehousing is a process which needs to occur before any data mining can take place.
Data mining is the considered as a process of extracting data from large data sets.	On the other hand, Data warehousing is the process of pooling all relevant data together.
One of the most important benefits of data mining techniques is the detection and identification of errors in the system.	One of the pros of Data Warehouse is its ability to update consistently. That's why it is ideal for the business owner who wants the best and latest features.
Data mining helps to create suggestive patterns of important factors. Like the buying habits of customers, products, sales. So that, companies can make the necessary adjustments in operation and production.	Data Warehouse adds an extra value to operational business systems like CRM systems when the warehouse is integrated.
The Data mining techniques are never 100% accurate and may cause serious consequences in certain conditions.	In the data warehouse, there is great chance that the data which was required for analysis by the organization may not be integrated into the warehouse. It can easily lead to loss of information.
The information gathered based on Data Mining by organizations can be misused against a group of people.	Data warehouses are created for a huge IT project. Therefore, it involves high maintenance system which can impact the revenue of medium to small-scale organizations.
After successful initial queries, users may ask more complicated queries which would increase the workload.	Data Warehouse is complicated to implement and maintain.

Data Mining	Data Warehouse
Organizations can benefit from this analytical tool by equipping pertinent and usable knowledge-based information.	Data warehouse stores a large amount of historical data which helps users to analyze different time periods and trends for making future predictions.
Organizations need to spend lots of their resources for training and Implementation purpose. Moreover, data mining tools work in different manners due to different algorithms employed in their design.	In Data warehouse, data is pooled from multiple sources. The data needs to be cleaned and transformed. This could be a challenge.
The data mining methods are cost-effective and efficient compares to other statistical data applications.	Data warehouse's responsibility is to simplify every type of business data. Most of the work that will be done on user's part is inputting the raw data.
Another critical benefit of data mining techniques is the identification of errors which can lead to losses. Generated data could be used to detect a drop-in sale.	Data warehouse allows users to access critical data from the number of sources in a single place. Therefore, it saves user's time of retrieving data from multiple sources.
Data mining helps to generate actionable strategies built on data insights.	Once you input any information into Data warehouse system, you will unlikely to lose track of this data again. You need to conduct a quick search, helps you to find the right statistic information.

Check Your Progress 1:

- 1) What are the descriptive and predictive data mining techniques? Explain.

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7.7 DATA MINING TOOLS

Data mining techniques are derived that utilize the domain knowledge from statistical analysis, artificial intelligence, and database systems in order to analyze data in a proper manner in view of different dimensions and perspectives.

Data mining tools discover patterns or trends from the collection of large sets of data and transforming data into useful information for making decisions.

These are the most popular data mining tools are Orange, SaS, Rattle Data Mining, Rapidminer, DataMelt Data Mining, Oracle BI etc..

7.7.1 Orange Data Mining

Developed at the bioinformatics laboratory at the faculty of computer and information science, Ljubljana University, Slovenia, Orange is a machine learning and data mining software suite. It supports data visualization and is a software-based on components written in Python. The components of Orange are called “widgets”.

These widgets range from data visualization and preprocessing to assessing the algorithms and are used for predictive modeling.

Widgets deliver much important functionalities such as:

- Displaying data table and allowing to select features
- Reading of data.
- Training predictors and comparison of learning algorithms
- Data element visualization, etc.

Data coming to orange is formatted quickly to the desired pattern, and the widgets can be easily transferred wherever and whenever needed. Orange is quite interesting to users. Orange allows its users to make smarter decisions in a short time by rapidly comparing and analyzing the data. Data mining can be performed via visual programming or Python scripting. Many analysis are feasible through its visual programming interface (drag and drop connected with widgets) and many visual tools tend to be supported such as bar charts, scatter-plots, trees, dendrograms, and heat maps. A substantial amount of widgets (more than 100) tend to be supported.

The instrument has machine learning components, add-ons for bioinformatics and text mining, and it is packed with features for data analytics. This is also used as a python library.

Orange comprises of canvas interface onto which the user places widgets and creates a data analysis workflow. The widget proposes fundamental operations, For example, reading the data, showing a data table, selecting features, training predictors, comparing learning algorithms, visualizing data elements, etc. Orange comes with multiple regression and classification algorithms.

Orange can read documents in native and other data formats. Orange is dedicated to machine learning techniques for classification or supervised data mining. There are two types of objects used in classification: learner and classifiers. Learners consider class-level data and return a classifier. Regression methods are very similar to classification in Orange, and both are designed for supervised data mining and require class-level data. The learning of ensembles combines the predictions of individual models for precision gain. The model can either come from different training data or use different learners on the same sets of data.

7.7.2 SAS Data Mining

SAS stands for Statistical Analysis System. It is a product of the SAS Institute created for analytics and data management. SAS can mine data, change it, manage information from various sources, and analyze statistics. It offers a graphical UI for non-technical users. SAS data miner allows users to analyze big data and provide accurate insight for timely decision-making purposes. SAS has distributed memory processing architecture that is highly scalable. It is suitable for data mining, optimization, and text mining purposes.

7.7.3 Rattle Data Mining

Rattle is a popular GUI for data mining using R. It presents statistical and visual summaries of data, transforms data so that it can be readily modelled, builds both unsupervised and supervised machine learning models from the data, presents the performance of models graphically, and scores new datasets for deployment into

production. Rattle is Free Open Source Software. Key feature is that all of your interactions through the graphical user interface are captured as an R script that can be readily executed in R independently of the Rattle interface. Use it as a tool to learn and develop your skills in R and then to build your initial models in Rattle to then be tuned in R which provides considerably more powerful options.

7.7.4 Rapid Miner

Rapid Miner is one of the most popular tools for performing predictions. It is written in JAVA programming language. It offers an integrated environment for text mining, deep learning, machine learning, and predictive analysis.

The instrument can be used for a wide range of applications, including company applications, commercial applications, research, education, training, application development, machine learning.

Rapid Miner provides the server on-site as well as in public or private cloud infrastructure. It has a client/server model as its base. A rapid miner comes with template-based frameworks that enable fast delivery with few errors(which are commonly expected in the manual coding writing process)

7.7.5 Data Melt

DataMelt is a free to use tool for numeric computation, mathematics, data analysis, and data visualization. This program offers you the simplicity of scripting languages, like Python, Ruby, Groovy with the power of hundreds of Java packages.

Features:

- DataMelt offers statistics, analysis of large data volumes, and scientific visualization.
- You can use it with different programming languages on different operating systems.
- It allows you to create high-quality vector-graphics images (EPS, SVG, PDF, etc.), which can be included in LaTeX and another text processor.
- Data Melt offers the usage of scripting languages, which are significantly faster than the standard Python implemented in C.

7.8 APPLICATIONS OF DATA MINING

Data Mining is primarily used today by companies with a strong consumer focus, in the areas like retail, financial, communication, and marketing organizations, to “drill down” into their transactional data and determine pricing, customer preferences and product positioning, impact on sales, customer satisfaction and corporate profits. With data mining, a retailer can use point-of-sale records of customer purchases to develop products and promotions to appeal to specific customer segments. Following are some of the applications of data mining:

Basket Analysis

In its most basic application, retailers use basket analysis to analyze what consumers buy (or put in their “baskets”). This is a form of the association technique, giving retailers insight into buying habits and allowing them to recommend other purchases. A less familiar application is one used by law enforcement, where vast amounts of

anonymous consumer data is analyzed looking for combinations of products one would use in bomb-making or the production of methamphetamine.

Sales Forecasting

Sales forecasting is a form of predictive analysis to which businesses are devoting more of their budgets. Data mining can help businesses project sales and set targets by examining historical data such as sales records, financial indicators (e.g., consumer price index, inflation markers etc.), consumer spending habits, sales attributed to a specific time of year, and trends which may impact standard assumptions about the business.

Bioinformatics

Data Mining approaches seem ideally suited for Bioinformatics, since it is data-rich. Mining biological data helps to extract useful knowledge from massive datasets gathered in biology, and in other related life sciences areas such as medicine and neuroscience. Applications of data mining to bioinformatics include gene finding, protein function inference, disease diagnosis, disease prognosis, disease treatment optimization, protein and gene interaction network reconstruction, data cleansing, and protein sub-cellular location prediction.

Inventory Planning

Data mining can provide businesses with up-to-date information regarding product inventory, delivery schedules, and production requirements. Data mining also can help remove some of the uncertainty that comes with simple supply-and-demand issues within the supply chain. The speed with which data mining can discern patterns and devise projections helps companies better manage their product stock and operate more efficiently.

Customer Segmentation

Traditional market research may help us to segment customers but data mining goes in deep and increases market effectiveness. Data mining aids in aligning the customers into a distinct segment and can tailor the needs according to the customers. Market is always about retaining the customers. Data mining allows to find a segment of customers based on vulnerability and the business could offer them with special offers and enhance satisfaction.

Customer Relationship Management

Customer Relationship Management is all about acquiring and retaining customers, also improving customers' loyalty and implementing customer focused strategies. To maintain a proper relationship with a customer a business need to collect data and analyze the information. This is where data mining plays its part. With data mining technologies the collected data can be used for analysis. Instead of being confused where to focus to retain customer, the seekers for the solution get filtered results.

Healthcare

Mining can be used to predict the volume of patients in every category. Processes are developed that make sure that the patients receive appropriate care at the right place and at the right time. Data mining can also help healthcare insurers to detect fraud and abuse.

Education

There is a new emerging field, called Educational Data Mining (EDM), concerns with developing methods that discover knowledge from data originating from educational Environments. The goals of EDM are identified as predicting students' future learning behavior, studying the effects of educational support, and advancing scientific knowledge about learning. Data mining can be used by an institution to take accurate decisions and also to predict the results of the student. With the results the institution can focus on what to teach and how to teach. Learning pattern of the students can be captured and used to develop techniques to teach them.

Intrusion Detection

Any action that will compromise the integrity and confidentiality of a resource is an intrusion. The defensive measures to avoid an intrusion includes user authentication, avoid programming errors, and information protection. Data mining can help improve intrusion detection by adding a level of focus to anomaly detection. It helps an analyst to distinguish an activity from common everyday network activity. Data mining also helps extract data which is more relevant to the problem.

Criminal Investigation

Criminology is a process that aims to identify crime characteristics. Actually crime analysis includes exploring and detecting crimes and their relationships with criminals. The high volume of crime datasets and also the complexity of relationships between these kinds of data have made criminology an appropriate field for applying data mining techniques. Text based crime reports can be converted into word processing files. This information can be used to perform crime matching process.

Fraud Detection

While frequently occurring patterns in data can provide teams with valuable insight, observing data anomalies is also beneficial, assisting companies in detecting fraud. While this is a well-known use case within banking and other financial institutions, SaaS-based companies have also started to adopt these practices to eliminate fake user accounts from their datasets.

Operational Optimization

Process mining leverages data mining techniques to reduce costs across operational functions, enabling organizations to run more efficiently. This practice has helped to identify costly bottlenecks and improve decision-making among business leaders.

7.9 ISSUES IN DATA MINING

Data mining is not an easy task, as the algorithms used can get very complex and data is not always available at one place. It needs to be integrated from various heterogeneous data sources. These factors also create some issues. Here in this tutorial, we will discuss the major issues regarding:

- Mining Methodology and User Interaction
- Performance Issues
- Diverse Data Types Issues

The following Figure 2 describes the major issues.

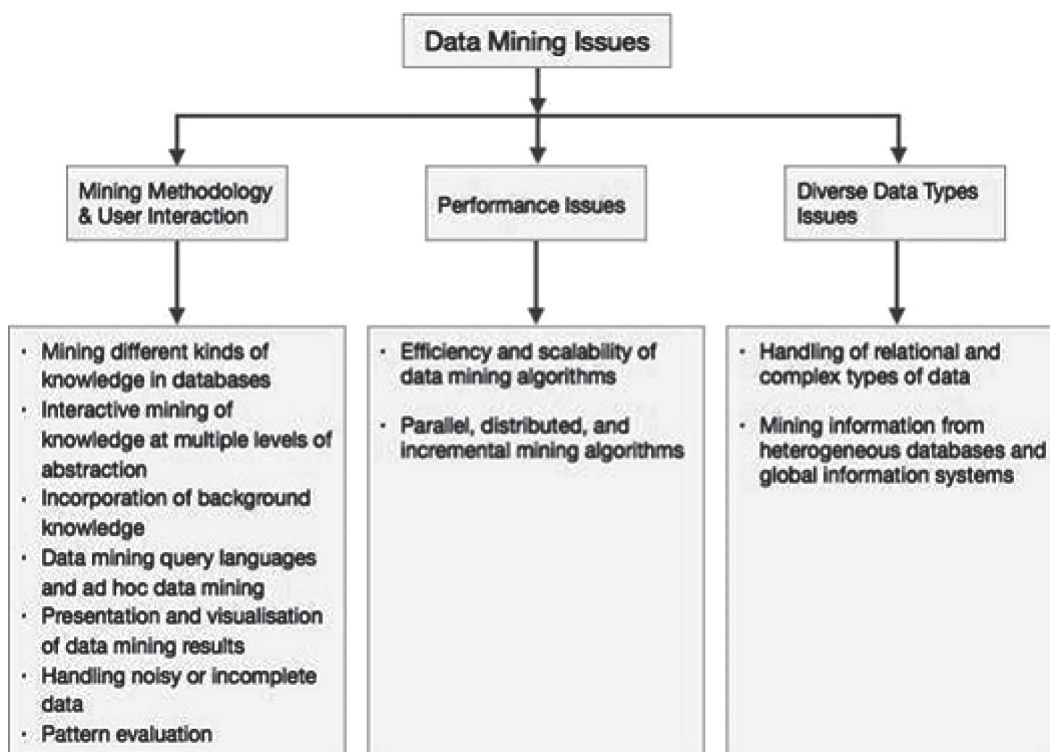


Figure 2: Mining Methodology and User Interaction Issues

7.9.1 Issues in Mining Methodology and User Interaction

It refers to the following kinds of issues –

- Mining different kinds of knowledge in databases – Different users may be interested in different kinds of knowledge. Therefore it is necessary for data mining to cover a broad range of knowledge discovery task.
- Interactive mining of knowledge at multiple levels of abstraction – The data mining process needs to be interactive because it allows users to focus the search for patterns, providing and refining data mining requests based on the returned results.
- Incorporation of background knowledge – To guide discovery process and to express the discovered patterns, the background knowledge can be used. Background knowledge may be used to express the discovered patterns not only in concise terms but at multiple levels of abstraction.
- Data mining query languages and ad hoc data mining – Data Mining Query language that allows the user to describe ad hoc mining tasks, should be integrated with a data warehouse query language and optimized for efficient and flexible data mining.
- Presentation and visualization of data mining results – Once the patterns are discovered it needs to be expressed in high level languages, and visual representations. These representations should be easily understandable.
- Handling noisy or incomplete data – The data cleaning methods are required to handle the noise and incomplete objects while mining the data regularities. If the data cleaning methods are not there then the accuracy of the discovered patterns will be poor.

- Pattern evaluation – The patterns discovered should be interesting because either they represent common knowledge or lack novelty.

7.9.2 Performance Issues

There can be performance-related issues such as –

- Efficiency and scalability of data mining algorithms – In order to effectively extract the information from huge amount of data in databases, data mining algorithm must be efficient and scalable.
- Parallel, distributed, and incremental mining algorithms – The factors such as huge size of databases, wide distribution of data, and complexity of data mining methods motivate the development of parallel and distributed data mining algorithms. These algorithms divide the data into partitions which is further processed in a parallel fashion. Then the results from the partitions is merged. The incremental algorithms, update databases without mining the data again from scratch.

7.9.3 Diverse Data Types Issues

- Handling of relational and complex types of data – The database may contain complex data objects, multimedia data objects, spatial data, temporal data etc. It is not possible for one system to mine all these kind of data.
- Mining information from heterogeneous databases and global information systems – The data is available at different data sources on LAN or WAN. These data source may be structured, semi structured or unstructured. Therefore mining the knowledge from them adds challenges to data mining.

Check Your Progress 2:

1. Identify and describe the features of some more open source / free data mining tools which were not discussed in this course content.

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7.10 SUMMARY

In this unit we had studied the introductory concepts of Data Mining which would be helpful for the learners to go through the rest of the units (Unit – 8 to Unit – 12) of this course.

Data mining is a technique that discovers previously unknown relationships in data. It is also known as Knowledge Discovery in Data (KDD). It is the practice of automatically searching large stores of data to discover patterns and trends that go beyond simple analysis. It uses sophisticated mathematical algorithms to segment the data and to predict the likelihood of future events based on past events. The key properties of data mining are namely - automatic discovery of patterns, prediction of likely outcomes, creation of actionable information and focus on large data sets and databases. Data mining can answer questions that cannot be addressed through simple query and reporting techniques.

Also, in this unit we had studied the applications of data mining and some data mining tools.

7.11 SOLUTIONS/ANSWERS

Check Your Progress 1:

1. Data mining encompasses a wide range of techniques and practices, but we can essentially sort them into two main types: descriptive and predictive.

Descriptive

Descriptive data mining techniques are used to determine the similarities in data and to identify patterns. Examples include:

(a) Association

This function is used to find interesting relationships and associations (hence the name) between items or values within datasets. For instance, it may be beneficial to know if certain products are often purchased together, as these items could be placed closer together in physical stores or offered as promotional packages in digital marketplaces.

(b) Clustering

Cluster analysis is used to group together items into clusters that share common characteristics. This technique can be applied to everything from biology to climate science to psychology. In business, clustering can be used to segment customers into small groups who may be receptive to particular marketing activities.

Predictive

Predictive data mining techniques are used to model future results using identified variables from the present. Examples include:

- (a) Classification: Classification generally involves a machine learning model which assigns items in a collection to predefined categories or classes. This may sound like a descriptive function, but the goal of classification is often to predict particular outcomes based on existing data. A classification model could, for instance, be used to identify loan applicants as low, medium, or high credit risks.
- (b) Regression: Regression is a statistical technique often employed in supervised machine learning that is used to (i) determine the relationship between a dependent variable and independent variables and (ii) use that relationship to predict a range of numeric values, given a particular dataset. Regression can, for instance, be used to predict the cost of a product or service when variables like the cost of fuel are considered.

Check Your Progress 2:

KNIME

KNIME is open source software for creating data science applications and services. This Data mining tool helps you to understand data and to design data science workflows.

Features:

- Helps you to build an end to end data science workflows
- Blend data from any source

- Allows you to aggregate, sort, filter, and join data either on your local machine, in-database or in distributed big data environments.
- Build machine learning models for classification, regression, dimension reduction

TANAGRA

Tanagra is a free to use data mining tool for study and research purposes. It offers various data mining methods from statistical learning, data analysis, and machine learning.

Features:

- Offers easy to use data mining software for researcher and students
- It allows the user to add their data mining methods.

ELKI

ELKI is an open source data mining tool written in Java. The tool allows us researching algorithms, with an emphasis on unsupervised methods in cluster analysis and outlier detection.

Features:

- ELKI offers an extensive collection of highly parameterizable algorithms.
- It allows easy and fair evaluation and benchmarking of algorithms.
- ELKI provides data index structures such as the R*-tree which enhance the process of Data mining

SPMF

SPMF is an open-source data mining library written in Java. It is distributed under the GPL license. It allows you to integrate source code with other Java Software.

Features:

- Allows association rule mining
- Supports sequential pattern and sequential rule mining
- Offers High-utility pattern mining,
- Time-series mining.
- Support complex process of Clustering and classification

Analytic Solver

Analytic Solver is free to use the point-and-click tool. It allows you to do risk analysis and prescriptive analytics in your browser. It offers full-power Data mining jobs.

Features:

- Helps you to incorporate uncertainty and solve with simulation optimization, stochastic programming, and robust optimization.
- Allows you to define the Monte Carlo simulation model using Excel formulae.

7.12 FURTHER READINGS

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